- **Drop:** Drops all incoming traffic unless related to outgoing traffic.

Managing firewalld

Firewalld can be managed in three ways.

- Using the command line tool: *firewall-cmd*
- Using the graphical tool: *firewall-confi*
- Using the configuration files in */etc/firewalld*

In most cases, editing the configuration files directly is not recommended, but it can be useful to copy information when using configuration management tools.

Configuring firewall settings with *firewall-cmd*

Firewall-cmd is installed as a part of the firewalld package, and can perform the same action as *firewall-config*.

The list below gives a number of frequently used *firewall-cmd* commands, along with explanations. Note that almost all commands will work on runtime configuration unless the *--permanent* option is specified. Many of the commands listed take the *--zone=<ZONE>* option to determine which zone they affect. If the *–zone* option is omitted from these commands, the default zone is used.

While configuring a firewall, an administrator normally applies all the changes permanently by using *firewall-cmd -- permanent --add service=service name* and then activates those changes with *firewall-cmd –reload*. While testing out new and possibly dangerous rules, an administrator can choose to work on the runtime configuration by omitting the *--permanent* option. In those cases, an extra option *--timeout=<TIMEINSECONDS>* can be used to automatically remove a rule after a certain amount of time, preventing an administrator from accidentally locking out a system.

Commands with examples

- *--get-default-zone:* Queries the current default zone.
- *--set-default-zone=<ZONE>:* Sets the default zone. This changes both the runtime and the permanent configuration.
- *--get-zones:* Lists all available zones.
- *--get-services:* Lists all predefined services.
- *--get-active-zones:* Lists all zones currently in use (and have an interface tied to them) along with their interface and source information.
- *--add-source=<CIDR> [--zone=<ZONE>]:* Routes all traffic coming from the IP address or network/netmask *<CIDR>* from the specified zone. If no *–zone=* option is provided, then the default zone will be used.
- *--remove-source=<CIDR>[--zone=<ZONE>]*: Removes the rule routing all traffic coming from the IP address or network/netmask *<CIDR>* from the specified zone. If no *–zone=* option is provided, the default zone will be used.
- *--add-interface=<INTERFACE> [--zone=<ZONE>]*: Routes all traffic coming from *<INTERFACE>* to the specified zone. If no *–zone=* option is provided, the default zone will be used.
- *--change-interface=<INTERFACE>[--zone=<ZONE>]:* Associates the interface with *<ZONE>* instead of its current zone. If no *–zone=* option is provided, the default zone will be used.
- *--list-all-zones:* Retrieves all information for all zones (interfaces, sources, ports, services, etc).
- *--add-service=<SERVICE>:* Allows traffic to *<SERVICE>*. If no *–zone=* option is provided, the default zone will be used.
- *--add-port=<PORT/PROTOCOL>*: Allows traffic to the *<PORT/PROTOCOL>* ports. If no *–zone=* option is provided, the default zone will be used.
- *--remove-service=<SERVICE>:* Removes *<SERVICE>* from the allowed list for the zone. If no *–zone=* option is provided, the default zone will be used.
- *--remove-port=<PORT/PROTOCOL>*: Removes the *<PORT/PROTOCOL>* ports from the allowed list for the zone. If no *–zone=* option is provided, the default zone will be used.
- *--reload:* Drops the runtime configuration and applies the persistent configuration.

Firewall-cmd shows the default zone being set to DMZ, all traffic coming from the *192.168.0.0/24* network being assigned to the internal zone, and network ports for MySQL being opened on the internal zone.

Firewalld configuration files

Firewalld configuration files are stored in two places -- */etc/firewalld* and */usr/lib/firewalld*. If a configuration file with the same name is stored in both locations, the version from */etc/firewalld* will be wiped out by a package update.

Managing rich rules

Apart from regular zones and the services syntax that firewalld offers, administrators have two other options for adding firewall rules — direct rules and rich rules.

Direct rules: Direct rules allow an administrator to insert hand-coded {ip, ip6, eb} tables rules into the zones managed by firewalld. While being powerful and exposing features of the kernel net filter sub-system that are not exposed through other means, these rules can be hard to manage. Direct rules also offer less flexibility than standard and rich rules. Unless explicitly inserted into a zone managed by firewalld, direct rules will be parsed before any firewalld rules are.

Shown below is an example of adding some direct rules to blacklist an IP range:

```
# firewall-cmd –direct --permanent –add-chain ipv4 raw
blacklist
# firewall-cmd –direct –permanent –add-rule ipv4 raw
PREROUTING 0 –s 192.168.0.0/24 –j blacklist
#firewall-cmd –direct –permanent –add-rule ipv4 raw blacklist
```